

Beam-only model: $m_{\text{ring}} \propto n \sqrt{\sin(\pi/n)}$ — this function is monotonically increasing. At $n=12$, beam mass is $2.19\times$ that of $n=3$. The naive conclusion: fewer lines = lighter rings. **But the beam-only model ignores the coupled system.** Thinner beams (95 vs 118 mm O